CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 3 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. An electric dipole placed in a uniform external electric field experiences: (a) a net force and a torque. (b) a net force but no torque. (c) a torque but strictly zero net force. (d) neither a force nor a torque. [PYQ 2020] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. The electric potential V at any point (x, y, z) in space is given by $V = 3x^2$, where x is in metres and V is in volts. The magnitude of the electric field at the point $(1\ \mathrm{m}, 0.2\ \mathrm{m}, 0\ \mathrm{m})$ is: (a) $6\ \mathrm{V/m}$ (b) $1.5\ \mathrm{V/m}$ (c) $3\ \mathrm{V/m}$ (d) Zero [PYQ 2021] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q3. In a current-carrying conductor, the ratio of the magnitude of the current density to the applied electric field is specifically defined as its: (a) Resistivity (b) Conductivity (c) Resistance (d) Mobility [PYQ 2020] | **[Chapter 3: Current Electricity | Confidence: High]**

Q4. Two long straight parallel wires carry steady currents in strictly opposite directions. The magnetic force between them will: (a) be attractive. (b) be repulsive. (c) be strictly zero. (d) alternate between attractive and repulsive. [PYQ 2021] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q5. A straight current-carrying wire kept in a uniform external magnetic field will experience a maximum magnetic Lorentz force when it is: (a) perpendicular to the magnetic field. (b) parallel to the magnetic field. (c) at an angle of 45° to the magnetic field. (d) at an angle of 60° to the magnetic field. [PYQ 2021] | **[Chapter 4: Moving Charges & Magnetism | Confidence: High]**

Q6. The Quality factor (Q-factor) of a series LCR resonant circuit is given by the mathematical expression: (a) $\frac{R}{L}\sqrt{\frac{L}{C}}$ (b) $\frac{1}{R}\sqrt{\frac{L}{C}}$ (c) $\frac{1}{R}\sqrt{\frac{C}{L}}$ (d) $\frac{1}{L}\sqrt{\frac{R}{C}}$ [Practice] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q7. Which of the following electromagnetic waves is primarily used as a diagnostic tool in modern medicine for imaging bones? (a) Gamma rays (b) Ultraviolet rays (c) X-rays (d) Infrared rays [PYQ 2020] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q8. Two thin lenses of optical powers $+5\ \mathrm{D}$ and $-2\ \mathrm{D}$ are placed coaxially in direct physical contact. The equivalent focal length of this lens combination is: (a) $+33.3\ \mathrm{cm}$ (b) $-33.3\ \mathrm{cm}$ (c) $+50.0\ \mathrm{cm}$ (d) $-50.0\ \mathrm{cm}$ [Practice] | **[Chapter 9: Ray Optics | Confidence: High]**

Q9. In a single slit diffraction experiment, if the wavelength of the incident monochromatic light is strictly increased, the angular width of the central maximum will: (a) Decrease (b) Increase (c) Remain entirely unchanged (d) Become zero [Practice] | **[Chapter 10: Wave Optics | Confidence: High]**

Q10. The graph showing the correct variation of the linear momentum (p) of a charged particle with its associated de-Broglie wavelength (λ) is a: (a) Straight line passing through the origin (b) Parabola (c) Rectangular hyperbola (d) Circle [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q11. The minimum energy required by an electron to just escape from the metal surface is fundamentally called the: (a) Kinetic energy (b) Threshold frequency (c) Stopping potential (d) Work function [Practice] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q12. A point charge $+Q$ is placed exactly at the geometrical centre of an uncharged, thick conducting spherical shell. The induced charge on the inner surface of this spherical shell will be: (a) $+Q$ (b) $-Q$ (c) Zero (d) $+Q/2$ [Practice] | **[Chapter 1: Electric Charges & Fields | Confidence: High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): A negative point charge released from rest in an electric field moves in a direction strictly opposite to the direction of the electric field. **Reason (R):** On a negative charge, the electrostatic force acts in the direction opposite to the electric field vector. [CBSE Term-1 2021] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): A step-up transformer can never be used as a step-down transformer under any circumstances. **Reason (R):** A transformer works only in one fixed direction of energy flow. [CBSE Term-1 2021] | **[Chapter 7: Alternating Current | Confidence: High]**

Q15. Assertion (A): In a purely inductive AC circuit, the alternating current lags behind the applied alternating voltage by exactly a phase angle of $\pi/2$. **Reason (R):** The inductive reactance of a coil increases directly proportionally with the frequency of the AC source. [Practice] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q16. Assertion (A): For incident radiation of a frequency strictly greater than the threshold frequency, the photoelectric current is directly proportional to the intensity of the incident radiation. **Reason (R):** Greater the number of energy quanta (photons) available per second, greater is the number of electrons absorbing these quanta and coming out of the metal. [CBSE SQP 2023] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. Two identical conducting balls A and B have electrical charges $-Q$ and $+3Q$ respectively. They are brought into physical contact with each other and then separated by a distance d in air. Find the exact nature and magnitude of the new Coulomb force between them in terms of Q and d. [PYQ 2019] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q18. Draw a clearly labelled graph showing the variation of electrical resistivity (ρ) of copper as a function of its absolute temperature (T). [PYQ 2014] | **[Chapter 3: Current Electricity | Confidence: High]**

Q19. A perfectly straight conducting wire of mass $200\ \mathrm{g}$ and length $1.5\ \mathrm{m}$ carries a steady current of $2\ \mathrm{A}$. It is suspended motionless in mid-air by a uniform horizontal magnetic field B. Calculate the magnitude of this magnetic field. (Take $g = 9.8\ \mathrm{m/s^2}$). [PYQ 2015] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q20. Why are infrared waves commonly referred to as 'heat waves'? State any one important practical application of infrared waves. [PYQ 2015] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q21. Two stationary point charges of $+1\ \mu\mathrm{C}$ and $+4\ \mu\mathrm{C}$ are kept exactly $30\ \mathrm{cm}$ apart in a vacuum. How far from the $+1\ \mu\mathrm{C}$ charge, precisely on the straight line joining the two charges, will the net electric field intensity be absolute zero? [PYQ 2020] | **[Chapter 1: Electric Charges & Fields | Confidence: High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Derive the mathematical expression for the electric field intensity at any general point on the equatorial line of an electric dipole of dipole moment $\vec{p}$. [PYQ 2019] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q23. A parallel plate capacitor of capacitance C is fully charged by a battery to a potential difference V. It is then disconnected from the battery and connected in parallel to another entirely uncharged capacitor of the exact same capacitance C. Calculate the precise ratio of the total electrostatic energy stored in the final combination to the initial energy stored in the single capacitor. [PYQ 2019] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q24. Define the term 'drift velocity' of charge carriers. Using the concept of drift velocity of free electrons in a metallic conductor, deduce the mathematical relationship between the macroscopic current density ($\vec{J}$) and the resistivity (ρ) of the conductor. [PYQ 2015] | **[Chapter 3: Current Electricity | Confidence: High]**

Q25. A straight metallic rod of length l is rotated with a constant angular velocity ω about an axis passing through one of its ends and perpendicular to the plane of rotation. A constant and uniform magnetic field B parallel to the axis of rotation is present everywhere. Using the concept of magnetic Lorentz force (or Faraday's laws), obtain the expression for the motional emf induced between the ends of the rod. [PYQ 2020] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q26. Draw a neat, labelled ray diagram showing the image formation by an astronomical refracting telescope in the normal adjustment position. Write the expression for its total magnifying power. [Practice] | **[Chapter 9: Ray Optics | Confidence: High]**

Q27. Monochromatic light of frequency $6.0 \times 10^{14}\ \mathrm{Hz}$ is produced by a laser source. The total optical power emitted is $2.0 \times 10^{-3}\ \mathrm{W}$. Calculate: (i) the exact energy of a single photon in the light beam (in Joules), and (ii) the total number of photons emitted on an average per second by the source. [PYQ 2018] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q28. An α-particle and a proton are accelerated from rest through the exact same accelerating potential difference V. Calculate the exact ratio of the de-Broglie wavelengths respectively associated with them. [PYQ 2017] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Alternating Current Resonance In a series LCR circuit connected to a variable-frequency AC source, the current amplitude strongly depends on the source frequency. When the frequency of the applied AC voltage perfectly matches the natural frequency of the LCR circuit, the current amplitude reaches its absolute maximum value. This phenomenon is known as electrical resonance. The specific frequency at which this occurs is called the resonant frequency (ω_r). At resonance, the inductive reactance completely cancels out the capacitive reactance. (A) What is the exact value of the power factor of a series LCR circuit at the condition of resonance? (1 Mark) (B) If the inductive reactance is strictly greater than the capacitive reactance ($X_L > X_C$) in the circuit, does the alternating current lead or lag behind the applied alternating voltage? (1 Mark) (C) A series LCR circuit consists of an inductor $L = 50\ \mathrm{mH}$, a capacitor $C = 80\ \mu\mathrm{F}$, and a resistor $R = 40\ \Omega$ connected to a $250\ \mathrm{V}$ variable frequency source. Determine the exact source angular frequency (ω_r) which drives the circuit into resonance. (2 Marks) [PYQ 2014 / Practice] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q30. Case Study: Young's Double Slit Experiment In 1801, Thomas Young brilliantly demonstrated the wave nature of light through his double-slit experiment. When monochromatic light from a single primary source illuminates two closely spaced narrow slits, the light waves emerging from these slits superimpose on a screen placed at a distance. This results in an interference pattern consisting of alternate bright and dark equally-spaced bands called fringes. For a sustained and stable interference pattern to be formed, the two secondary sources of light must be strictly coherent. (A) Define the term 'coherent sources' of light. (1 Mark) (B) How is the linear fringe width (β) on the screen explicitly affected if the screen is moved further away from the plane of the slits? (1 Mark) (C) In a standard Young's double-slit experiment, the two slits are made exactly $1.0\ \mathrm{mm}$ apart and the observation screen is placed $1.0\ \mathrm{m}$ away. What is the linear fringe separation (in mm) when a blue-green laser light of wavelength 500 nm is used? (2 Marks) [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) Based on the concept of bound charges and polarization, define a 'dielectric'. (B) Derive the mathematical expression for the capacitance of a parallel plate capacitor having plate area A and separation d, when a solid dielectric slab of thickness t ($t < d$) and dielectric constant K is partially inserted between its plates. (C) An equiconvex lens of focal length 20 cm in air is strictly made of glass of refractive index 1.5. Find its exact new focal length when it is completely immersed in a transparent liquid of refractive index 1.25. [PYQ 2022 / Practice] | **[Chapter 2: Electrostatic Potential & Capacitance / Chapter 9: Ray Optics | Confidence: Very High]** *(Note: Part C integrates a basic ray optics application, common in 5-mark multi-concept questions).*

Q32. (A) State the underlying principle of a Moving Coil Galvanometer. (B) With the help of a schematic diagram, derive the mathematical expression for the deflecting torque acting on the rectangular current-carrying coil of the galvanometer when placed in a uniform magnetic field. Hence, write the condition for steady angular deflection. (C) What is the specific physical significance of employing a 'radial magnetic field' in the moving coil galvanometer? How is it practically achieved? [PYQ 2019, 2020] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q33. (A) Draw a neat and clearly labelled ray diagram showing the refraction of a monochromatic ray of light through a spherical convex refracting surface separating two media of refractive indices n_1 and n_2 ($n_2 > n_1$), when the object is placed in the rarer medium. (B) Using this diagram, derive the fundamental refraction formula: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$. (C) Apply this derived relation successively to the two spherical surfaces of a thin convex lens to systematically deduce the Lens Maker's Formula: $\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. [Practice / PYQ] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Answer Key

Q1: (c) A torque but strictly zero net force. In a uniform field, the $+q$ and $-q$ forces cancel out, but their different lines of action produce a torque. **Q2:** (a) 6 V/m. $E = -dV/dx = -\frac{d}{dx}(3x^2) = -6x$. At $x = 1$, magnitude $|E| = 6(1) = 6 \text{ V/m}$. **Q3:** (b) Conductivity. By Ohm's law vector form, $J = \sigma E \Rightarrow \sigma = J/E$. **Q4:** (b) be repulsive. Parallel currents attract; anti-parallel (opposite) currents strictly repel each other due to magnetic force interactions. **Q5:** (a) perpendicular to the magnetic field. $F = IlB\sin\theta$. Max force occurs when $\theta = 90^\circ$ ($\sin 90^\circ = 1$). **Q6:** (b) $\frac{1}{R}\sqrt{\frac{L}{C}}$. This is the standard formula for the Q-factor in a series LCR circuit. **Q7:** (c) X-rays. X-rays can penetrate soft tissues and are widely used in medical imaging (radiography) for examining bones. **Q8:** (a) $+33.3 \text{ cm}$. Total Power $P = P_1 + P_2 = +5 - 2 = +3 \text{ D}$. Focal length $F = 1/P = 1/3 \text{ m} = 100/3 \text{ cm} = +33.3 \text{ cm}$. **Q9:** (b) Increase. The angular width is given by $2\lambda/a$. Since it is directly proportional to λ, an increase in wavelength increases the angular width. **Q10:** (c) Rectangular hyperbola. According to de-Broglie, $p = h/\lambda$. Since $p\lambda = \text{constant}$, the graph is a rectangular hyperbola. **Q11:** (d) Work function. It is defined as the minimum energy required to eject an electron from the metal surface. **Q12:** (b) $-Q$. By Gauss's Law, to keep the electric field inside the bulk of the conductor strictly zero, an equal and opposite charge $-Q$ must be induced on the inner surface.

Q13: (a) Both A and R are true and R is the correct explanation of A. **Q14:** (d) A is false and R is also false. A step-up transformer can easily be used as a step-down transformer simply by reversing the input and output terminals (primary and secondary roles). **Q15:** (b) Both A and R are true but R is NOT the correct explanation of A. The phase lag is due to the inherent property of an inductor opposing changes in current (Faraday's/Lenz's laws), not simply because $X_L \propto f$. **Q16:** (a) Both A and R are true and R is the correct explanation of A.

Q17: When in contact, net charge $= -Q + 3Q = +2Q$. It distributes equally, so each gets $+Q$. New force $F = \frac{1}{4\pi\epsilon_0} \frac{Q \times Q}{d^2} = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{d^2}$. Nature: Repulsive (both are positive). **Q18:** Plot a curve with T on the x-axis and ρ on the y-axis. The curve starts from a non-zero positive intercept at $T = 0$ and rises almost linearly for higher temperatures. **Q19:** Magnetic force balances weight: $IlB = mg$. $B = \frac{mg}{Il} = \frac{0.200 \times 9.8}{2 \times 1.5} = \frac{1.96}{3.0} \approx 0.653 \text{ T}$. **Q20:** They are called heat waves because they are readily absorbed by water molecules in most materials, which heavily increases the thermal motion of molecules, thereby heating up the material. Use: TV remotes, treating muscular strains, night vision. **Q21:** Let the null point be at distance x from $+1\ \mu\text{C}$. The electric fields must balance: $\frac{k(1)}{x^2} = \frac{k(4)}{(30-x)^2}$. Taking the square root: $\frac{1}{x} = \frac{2}{30-x} \Rightarrow 30 - x = 2x \Rightarrow 3x = 30 \Rightarrow x = 10 \text{ cm}$.

Q22: Draw an electric dipole with charges $+q$ and $-q$ separated by $2a$. Consider an equatorial point P at distance r. Find E_{+q} and E_{-q} (magnitudes are equal: $\frac{kq}{r^2+a^2}$). Resolve into components; the vertical components cancel, horizontal components add up. $E_{net} = 2E\cos\theta = 2\left(\frac{kq}{r^2+a^2}\right)\left(\frac{a}{\sqrt{r^2+a^2}}\right) = \frac{kp}{(r^2+a^2)^{3/2}}$. **Q23:** Initial energy $U_i = \frac{1}{2}CV^2$. When connected in parallel, total charge $Q = CV$ is shared equally. Common potential $V' = \frac{Q_{total}}{C_{total}} = \frac{CV}{2C} = \frac{V}{2}$. Final energy $U_f = \frac{1}{2}(2C)(V/2)^2 = \frac{1}{2}(2C)(V^2/4) = \frac{1}{4}CV^2$. Ratio $U_f/U_i = (\frac{1}{4}CV^2)/(\frac{1}{2}CV^2) = 1 : 2$. **Q24:** Drift velocity is the average velocity with which free electrons drift towards the positive terminal under an external electric field. $I = neAv_d$. We know $v_d = \frac{eE\tau}{m}$. Substituting, $I = neA(\frac{eE\tau}{m}) \Rightarrow \frac{I}{A} = \frac{ne^2\tau}{m}E$. Since $J = I/A$, $J = \left(\frac{ne^2\tau}{m}\right)E$. Comparing with $J = \sigma E = \frac{E}{\rho}$, we get $\rho = \frac{m}{ne^2\tau}$. **Q25:** Consider a small element dr at distance r from the axis. Its linear velocity is $v = r\omega$. The motional emf induced in this element is $de = Bv(dr) = B(r\omega)dr$. Total emf $\epsilon = \int_0^l B\omega r dr = B\omega\left[\frac{r^2}{2}\right]_0^l = \frac{1}{2}B\omega l^2$. **Q26:** Diagram must show two convex lenses

(Objective large, Eyepiece small). Parallel rays from infinity form an intermediate real inverted image at the common focus (F_o and F_e coincide). Final image forms at infinity. Magnifying power $M = -f_o/f_e$. **Q27:** (i) $E = h\nu = (6.63 \times 10^{-34}) \times (6.0 \times 10^{14}) \approx 3.98 \times 10^{-19}\ \mathrm{J}$. (ii) Number of photons $n = P/E = (2.0 \times 10^{-3})/(3.98 \times 10^{-19}) \approx 5.0 \times 10^{15}\ \mathrm{photons/s}$. **Q28:** de-Broglie wavelength $\lambda = \frac{h}{\sqrt{2mqV}}$. Ratio $\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p q_p V}{m_\alpha q_\alpha V}} = \sqrt{\frac{m_p \times e}{4m_p \times 2e}} = \sqrt{\frac{1}{8}} = \frac{1}{2\sqrt{2}}$.

Q29: (A) At resonance, $Z = R$, so Power Factor $\cos\phi = R/Z = R/R = 1$. (B) If $X_L > X_C$, the circuit is net-inductive, so the current strictly **lags** behind the applied voltage. (C) $\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 10^{-3} \times 80 \times 10^{-6}}} = \frac{1}{\sqrt{4000 \times 10^{-9}}} = \frac{1}{\sqrt{4 \times 10^{-6}}} = \frac{1}{2 \times 10^{-3}} = 500\ \mathrm{rad/s}$. **Q30:** (A) Coherent sources are two sources of light that continuously emit waves of the exact same frequency and maintain a constant (or zero) phase difference over time. (B) Fringe width $\beta = \lambda D/d$. If screen distance D increases, the fringe width β linearly increases. (C) $\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1.0}{1.0 \times 10^{-3}} = 500 \times 10^{-6}\ \mathrm{m} = 0.5\ \mathrm{mm}$.

Q31: (A) A dielectric is a non-conducting material where external electric fields cause microscopic localized displacement of bound charges, producing polarization. (B) Diagram of capacitor with slab. $V = E_0(d-t) + E_m(t)$, where $E_m = E_0/K$. $V = E_0(d - t + t/K) = \frac{Q}{A\epsilon_0}(d - t + t/K)$. $C = \frac{Q}{V} = \frac{\epsilon_0 A}{d - t(1 - 1/K)}$. (C) In air, $\frac{1}{20} = (1.5 - 1)(\frac{2}{R}) \Rightarrow \frac{1}{20} = \frac{0.5 \times 2}{R} \Rightarrow R = 20\ \mathrm{cm}$. In liquid, $\frac{1}{f_l} = (\frac{1.5}{1.25} - 1)(\frac{2}{20}) = (1.2 - 1)(0.1) = 0.2 \times 0.1 = 0.02 \Rightarrow f_l = 1/0.02 = 50\ \mathrm{cm}$.
Q32: (A) A current-carrying coil placed in a magnetic field experiences a torque. (B) Draw a rectangular coil ABCD. Forces on horizontal arms are collinear and cancel. Forces on vertical arms are $F = NIlB$, forming a couple. Torque $\tau = F \times \text{perpendicular distance} = NIlB \times b\sin\theta = NIAB\sin\theta$. Steady deflection condition: Restoring torque = Deflecting torque ⇒ $k\alpha = NIAB$. (C) A radial magnetic field ensures that the plane of the coil is always exactly parallel to the magnetic field ($\theta = 90^\circ, \sin 90^\circ = 1$) irrespective of its rotation, making the scale perfectly linear ($\alpha \propto I$). Achieved by using concave cylindrical pole pieces and a soft iron core. **Q33:** (A) Draw convex surface, object O on principal axis in rarer medium, refracted ray bending towards normal to form real image I. Show angles $i, r, \alpha, \beta, \gamma$. (B) For paraxial rays, $n_1 \sin i = n_2 \sin r \Rightarrow n_1 i = n_2 r$. Use geometry exterior angles $i = \alpha + \gamma$ and $\gamma = r + \beta \Rightarrow r = \gamma - \beta$. Substitute: $n_1(\alpha + \gamma) = n_2(\gamma - \beta) \Rightarrow n_1\alpha + n_2\beta = (n_2 - n_1)\gamma$. Using $\tan\theta \approx \theta$, substitute $\alpha \approx h/u, \beta \approx h/v, \gamma \approx h/R$ (with sign conventions $u \to -u$). Result: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$. (C) Apply to surface 1: image forms at v'. $\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1}$. Apply to surface 2 (denser to rarer): $\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2} = \frac{-(n_2 - n_1)}{R_2}$. Adding both: $n_1(\frac{1}{v} - \frac{1}{u}) = (n_2 - n_1)(\frac{1}{R_1} - \frac{1}{R_2})$. Dividing by n_1 and using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ yields the Lens Maker's Formula.
